arohak

Purpose. People. Innovation

QUANTUM-INSPIRED OPTIMISATION · 112 USE CASE

Faster emergency response, solved on an Ising machine

Ambulance dispatch modelled as a QUBO and solved with light

In partnership with **Quanfluence**

How it works from distress call to dispatch plan

01 — Capture the situation

Distress calls, ambulance locations and road travel times are gathered for every call.

02 — Model it as a QUBO

Every choice becomes a yes/no switch: *does ambulance A take call 1?* Travel time adds cost, and broken rules (one ambulance per call, each ambulance used once) add heavy penalties.

03 — Fit it to the hardware

The state-level problem is split by zone so each piece fits the Ising machine. Fitting the problem to the hardware is a big part of the engineering.

04 — Solve with light

On Quanfluence's Ising machine, each spin is a pulse of light in a fibre loop. The pulses interact and settle into the lowest-energy state: the best plan.

05 — Decode & dispatch

The spins are turned back into a dispatch plan: the right ambulance and the fastest route for every call in the zone.

BUILT TO EXTEND

The model can include rules such as matching the patient to the right hospital.

ROADMAP

✓ Simulation on 108 call data — 108 Live ambulance trial — 112 Police · Fire · Ambulance

Quantum-inspired today. Quantum-ready tomorrow.